MOHAMMAD R. AL-EIADEH

Ph.D. in Electrical and Computer Engineering

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SUMMARY

Mohammad Riyad Al-Eiadeh is a dedicated Ph.D. candidate in the Elmore Family School of Electrical and Computer Engineering at Purdue University. He earned his B.Sc. and M.Sc. in Computer Science from Al-Balqa Applied University in 2018 and Yarmouk University in 2021, respectively. His **research** focuses on **Graph Machine Learning, Graph Computations, Adversarial Machine Learning, Adversarial Attacks on Graph Embeddings and Graph Structures, Robust and Resilient Graph Learning, Matrix Perturbation Theory, Complex Networks, Dynamic Networks, Tensor Product of Graphs, Higher-Order Graphs, and Cybersecurity (Interdependent Systems modeled as Attack Graphs)**. He has experience in optimization problems such as graph embedding and scalable graph analytics. He is interested in learning on graphs, Java, multi-threading, functional programming, and data structures, and is passionate about solving complex challenges in computer science, AI security, and resilient intelligent systems. Contact him at maleiade@purdue.edu.

SKILLS

Languages: Java, C++, C#, Python.

Technologies: EJML (Matrix Library), Tribuo (ML library), DFLib (DataFrame library), JGraphT library, XChart library, Open CSV API, JDBC, Collection Framework, Stream & Parallel Stream, Executor Service, Maven.

Programming Paradigms: OOP, Packaging, Design Pattern, Generics, Functional | Declarative Programming.

Research Interests: Graph ML, Cyber Security, Matrix Computation, Network Science, Complex Networks & Time Complexity Analysis.

EDUCATION

8/2023 - 6/2026 **Purdue University, Indiana, USA** — **School**
Ph.D. in Electrical and Computer Engineering, 90 credit hours
Thesis Fields: Cybersecurity, Graph Algorithms, Attack Graphs, Resource Allocation, Machine Learning

9/2019 - 7/2022 **Yarmouk University, Irbid, Jordan** — **School**
Masters' in Computer Science, 32 Credit Hours
Thesis Fields: Feature Selection. Evolutionary Algorithm. Machine Learning

9/2015 - 7/2018 **Al-Huson University College | Al-Balqa Applied University, Irbid, Jordan** — **School**
Bachelors' in Computer Science, 132 Credit Hours
Project Fields: Android, Arduino, Smart Home

EXPERIENCE

8/2023 – Present **Graduate Researcher & Software Developer** — **Advisor: Dr. Mustafa Abdallah**

- **Securing Interdependent Systems**: Developing graph-theoretic and graph-learning-based models for protecting interdependent systems represented as attack graphs.

1/2026 – Present **Graduate Teaching Assistant** — **Courses: CIT 44000, CIT 40600 — Purdue University**

- Assisting in **Communication Network Design** & **Advanced Network Security**; grade homework and laboratory assignments, Provide academic support and hold office hours.

8/2025 – 12/2025 **Graduate Teaching Assistant** — **Course: CNIT 25501 — Purdue University**

- Assisted in **Java programming**, mentoring students in **OOP, basic GUI development, and problem solving**; evaluated coursework and provided technical support during lab sessions.

1/2025 – 5/2025 **Graduate Teaching Assistant** — **Purdue University**

- Mentored students in programming laboratories, assisting with **data structures, algorithms, and debugging**, and providing technical guidance during lab sessions.

8/2024 – 12/2024 **Graduate Teaching Assistant** — **Course: ENGR133 — Purdue University**

- Supported instruction in **Python and MATLAB** for first-year engineering students, helping develop computational thinking and engineering problem-solving skills.

8/2022 – 8/2023 **Researcher & Software Developer** **Advisor: Prof. Mohammad Batah**

- Developed hybrid **evolutionary optimization algorithms** for NP-hard combinatorial problems, including **Maximum Clique (DIMACS)**, **Feature Selection** & **0–1 Knapsack**.

AWARDS & HONORS

7/2023 **Purdue School of Engineering and Technology, IUPUI, IN, USA**
Graduate Research Assistant-ship, PhD in ECE at E & T Department

6/2018 **Al-Huson University College, Al-Huson, Jordan**
Academic Excellence Award - Ranked 1st in my class for academic performance

PHD COURSES

ECE 62900– INTRO NEURAL NETWORKS
ECE 66200– PATTN RECOG / DECIS PROC
MA 51100 – LINEAR ALGEBRA APPL
ECE 58000– OPT METH FOR SYS & CON
ECE 59500 – INTRODUCTION TO GAME THEORY
MA 57500 – GRAPH THEORY

PUBLICATIONS

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. "Auto-Sec-HS: Meta-Learning Approach Based on Hand-Crafted Domain-Based Features for Efficient Security Resource Allocation on Attack Graphs,"** ***Journal of Information Security and Applications (JISA)***, **Under Review, 2026.**

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. "A Robustness Analysis of Attack-Graph-Driven Security Resource Allocation Schemes under Adversarial Attacks,"** ***ACM Transactions on Privacy and Security (TOPS)***, **Under Review, 2026**

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. Auto-Sec: Meta-Learning for Automated Efficient Security Resource Allocation on Attack Graphs. IEEE Transactions on Dependable and Secure Computing, 2025.** GitHub

Conference Paper — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. SAG-ERC: Securing Attack-Graph-based Systems through Node Embedding, Ranking, and Clustering. IEEE Conference on Secure and Trustworthy CyberInfrastructure for IoT and Microelectronics (SaTC), Ohio, Dayton, USA, 2025.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems. ACM Trans. Priv. Secur., 2025.** GitHub

Journal Article — **Al-Batah, M. S., Mohammad Ryiad Al-Eiadeh, & Alnsour, Y. A Novel Approach for Enhancing Plant Leaf Classification With the Binary Cuckoo Search Algorithm. Applied Computational Intelligence and Soft Computing, 2025.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs. International Journal of Information Security, 2025.** GitHub

Conference Paper — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. AARA-PR: Asset-Aware PageRank-Based Security Resource Allocation Method for Attack Graphs. IEEE Interdisciplinary Conference on Electrics and Computer (INTCEC), IEEE, Chicago, IL, USA, 2024.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. Genigraph: A Genetic-Based Novel Security Defense Resource Allocation Method for Interdependent Systems Modeled by Attack Graphs. Computers & Security, 2024.** GitHub

Journal Article — **Al-Batah, M. S., & Mohammad Ryiad Al-Eiadeh. An Improved Binary Crow-JAYA Optimization System With Various Evolution Operators Such as Mutation for Finding the Max Clique in Dense Graphs. International Journal of Computing Science and Mathematics, 2024.** GitHub

Journal Article **Mohammad Ryiad Al-Eiadeh, Raneem Qaddoura, and Mustafa Abdallah. Investigating the Performance of a Novel Modified Binary Black Hole Optimization Algorithm for Enhancing Feature Selection. Applied Sciences, 2024.** GitHub

Journal Article **Al-Batah, M. S., & Mohammad Ryiad Al-Eiadeh. An Improved Discrete Jaya Optimisation Algorithm With Mutation Operator and Opposition-Based Learning to Solve the 0-1 Knapsack Problem. International Journal of Mathematics in Operational Research, 2023.** GitHub

Journal Article **Mohammad Ryiad Al-Eiadeh. Automatic Lung Field Segmentation Using Robust Deep Learning Criteria. International Journal of Hybrid Information Technologies, 2021.**

– POSTER PRESENTATIONS

2026 **CERIAS Annual Security Symposium, Purdue University** **Poster Presentation**
Mohammad Al-Eiadeh and Mustafa Abdallah. *CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems.* Poster

2025 **CERIAS Annual Security Symposium, Purdue University** **Poster Presentation**
Mohammad Al-Eiadeh and Mustafa Abdallah. *PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs.* Poster

PROFESSIONAL SERVICE

2026 **IEEE** **Journal Peer Reviewer**
Reviewed **1 manuscript** for *IEEE Transactions on Information Forensics and Security (TIFS).*

2026 **Springer Nature** **Journal Peer Reviewer**
Reviewed **1 manuscript** for *Journal of King Saud University Computer and Information Sciences.* Certificate

2026 **Springer Nature** **Journal Peer Reviewer**
Reviewed **1 manuscript** for *International Journal of Machine Learning and Cybernetics.* Certificate

2025 **Springer Nature** **Journal Peer Reviewer**
Reviewed **1 manuscript** for *Cluster Computing.* Certificate

2025 **Springer Nature** **Journal Peer Reviewer**
Reviewed **1 manuscript** for *Applied Network Science.* Certificate

REFERENCE

Scholar **Purdue University, Indianapolis & West Lafayette, IN, USA** **Supervisor**
Prof. Mustafa Abdallah
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Scholar **Purdue University, Indianapolis & West Lafayette, IN, USA** **Co-Supervisor**
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Prof. Maher Rizkalla
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Instructor **Purdue University, Indianapolis & West Lafayette, IN, USA** **Assistant & Cordinator**
Sherrie Tucker
Email: tucker76@purdue.edu

PROJECTS

Software Project **GraphEmbedding4j: In-progress Java library for graph / node embedding research, with modular components for random walks, context-window extraction, negative sampling, skip-gram learning, gradient descent optimization, and sigmoid-based loss modeling.** GitHub

LANGUAGES

Arabic - **Native**, **English** - **Fluent**